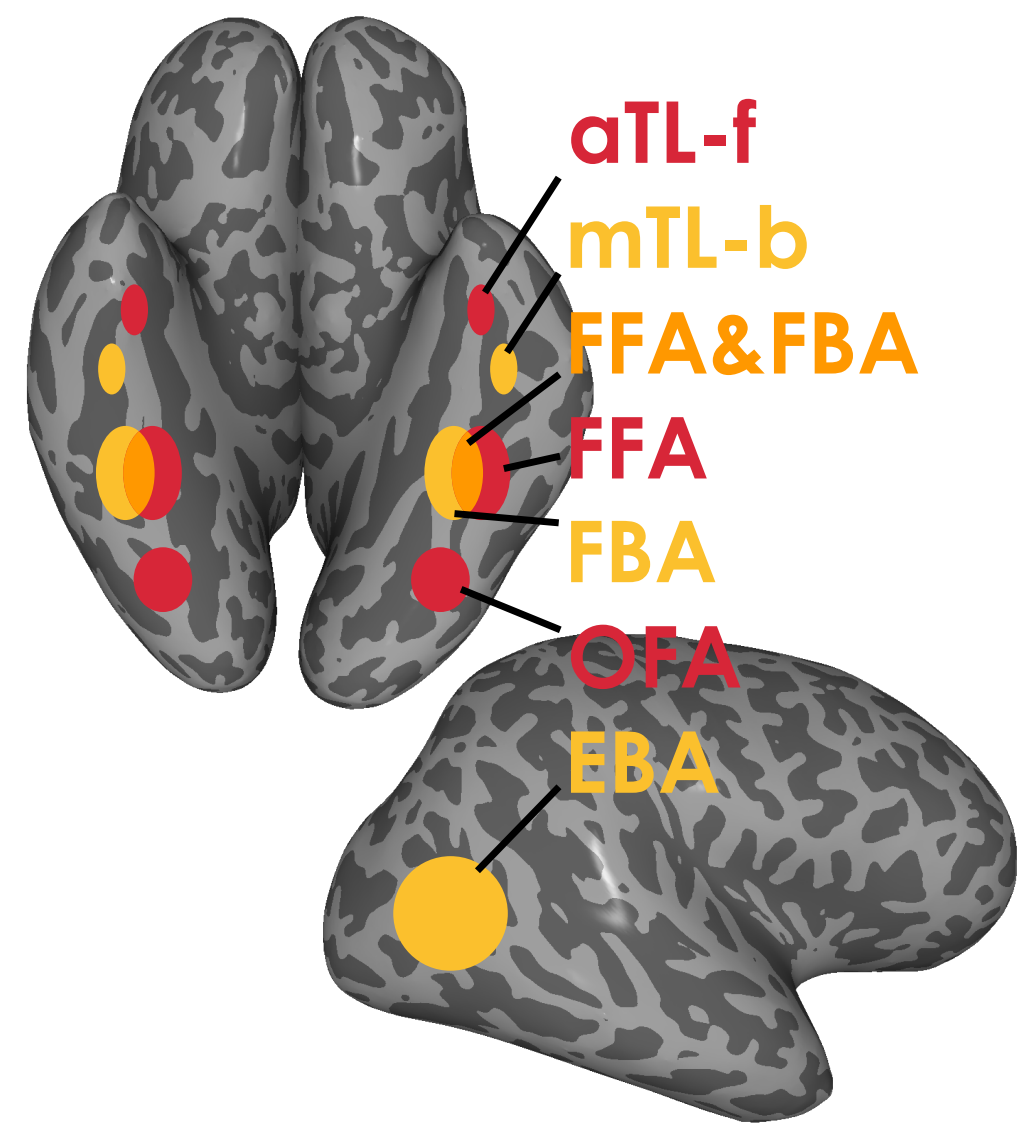


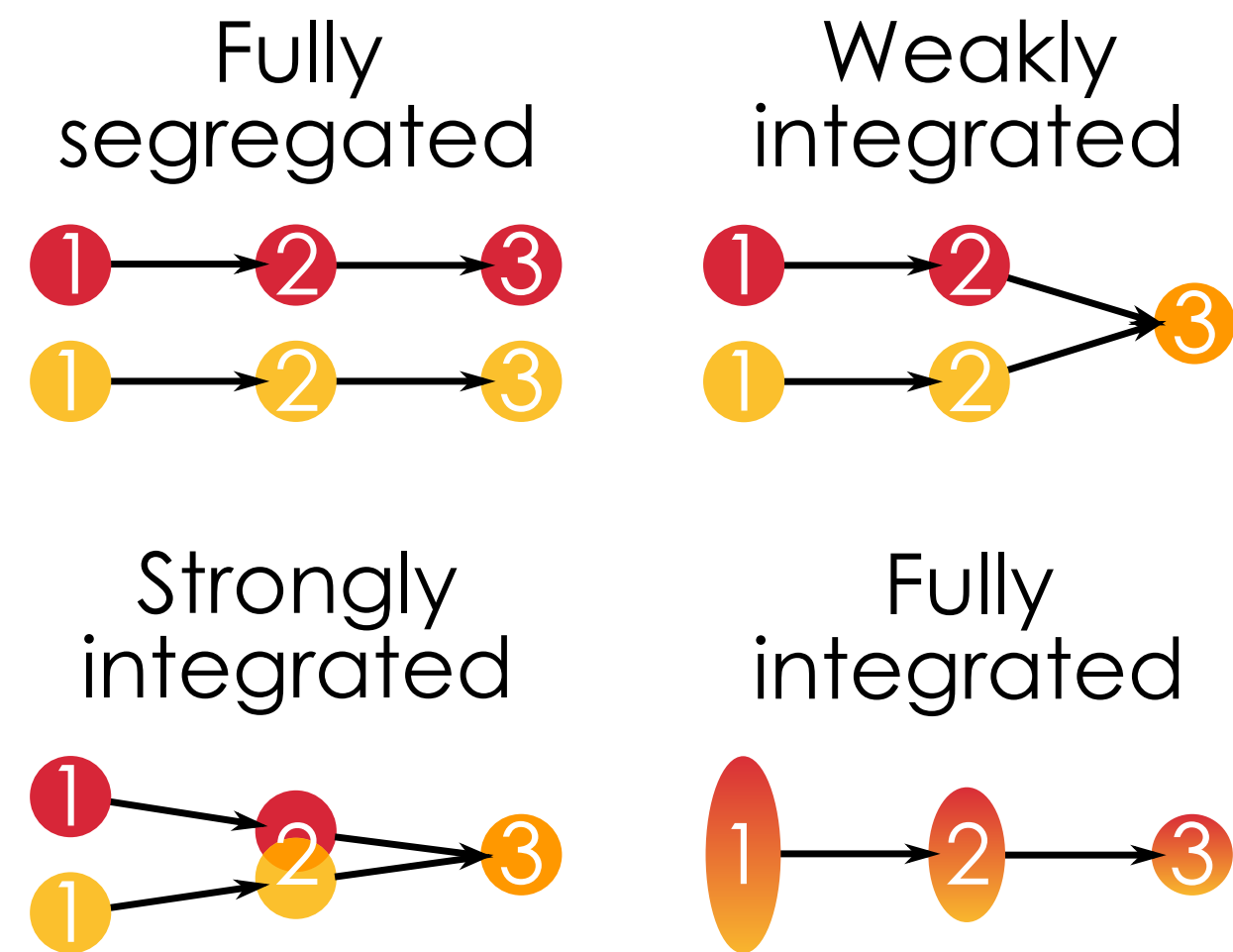
Deep neural networks provide insights into distinct and shared selectivity for faces and bodies in human visual cortex

How segregated vs. integrated are face and body processing in visual cortex?

Specialized areas¹
with substantial overlap²



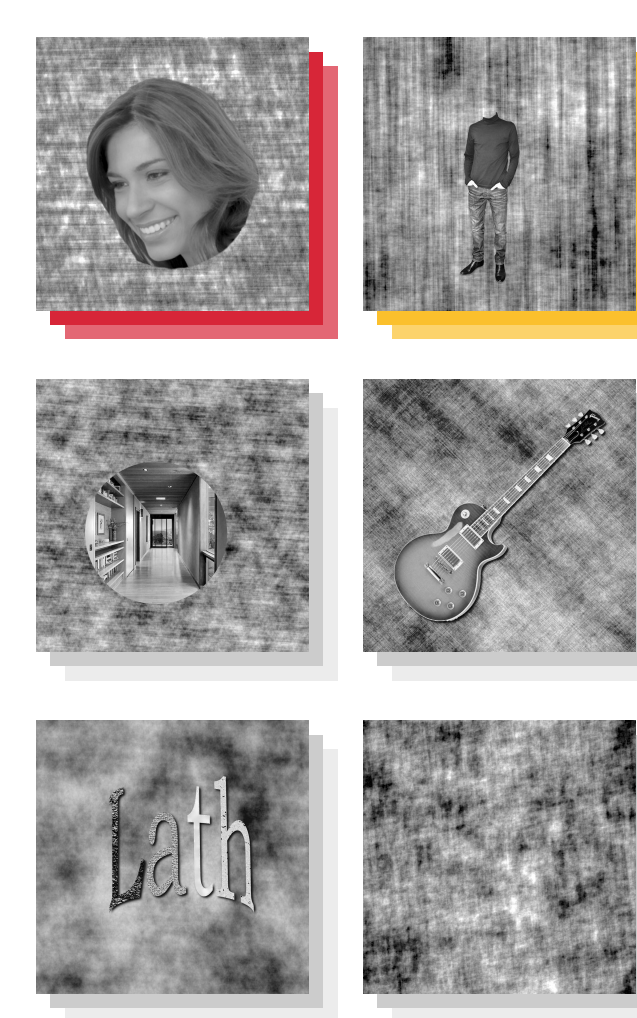
Competing
hypotheses³



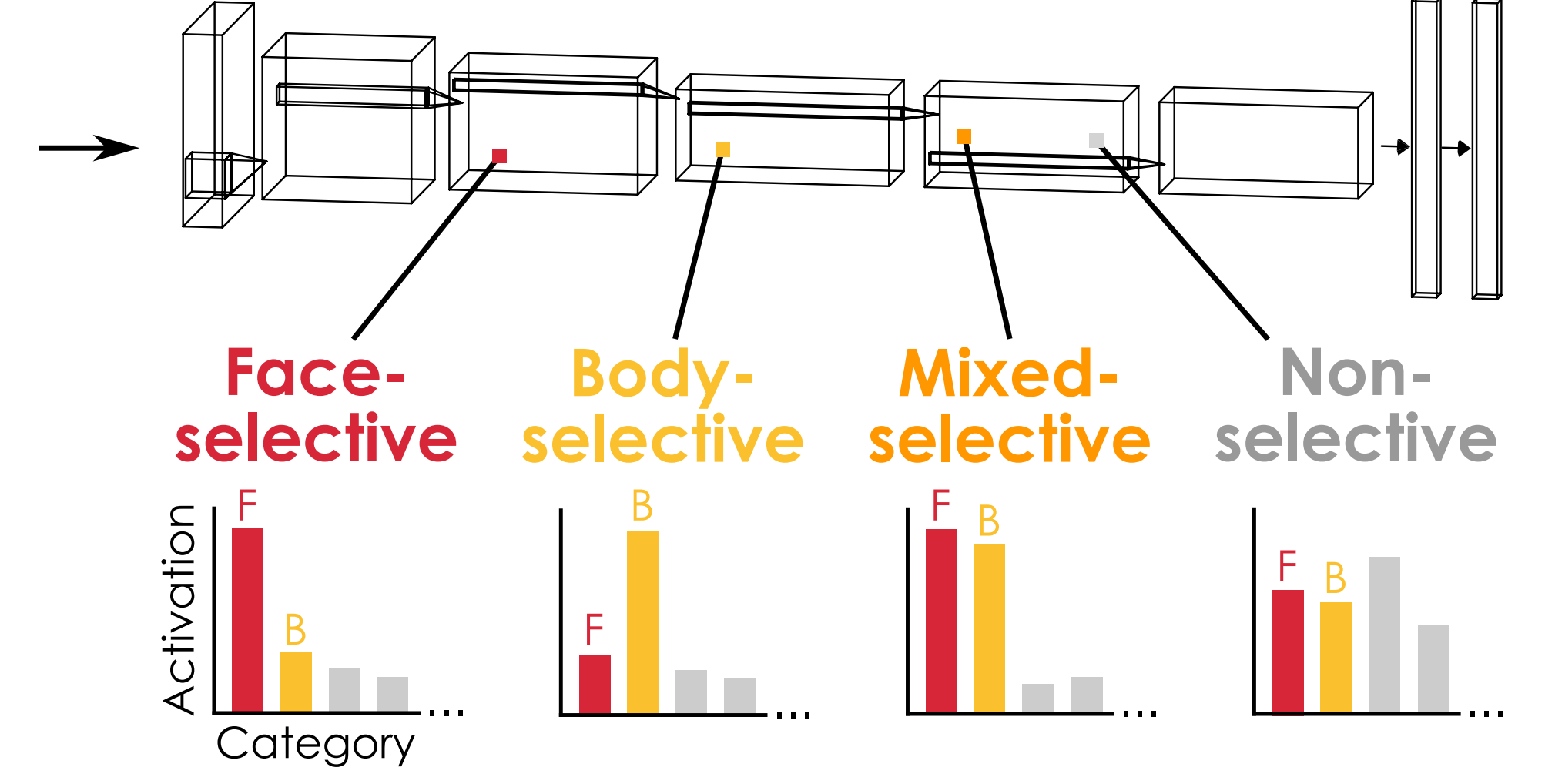
■ Face-selective ■ Body-selective ■ Mixed-selective

Testing competing hypotheses in DNNs

Functional localizer⁴



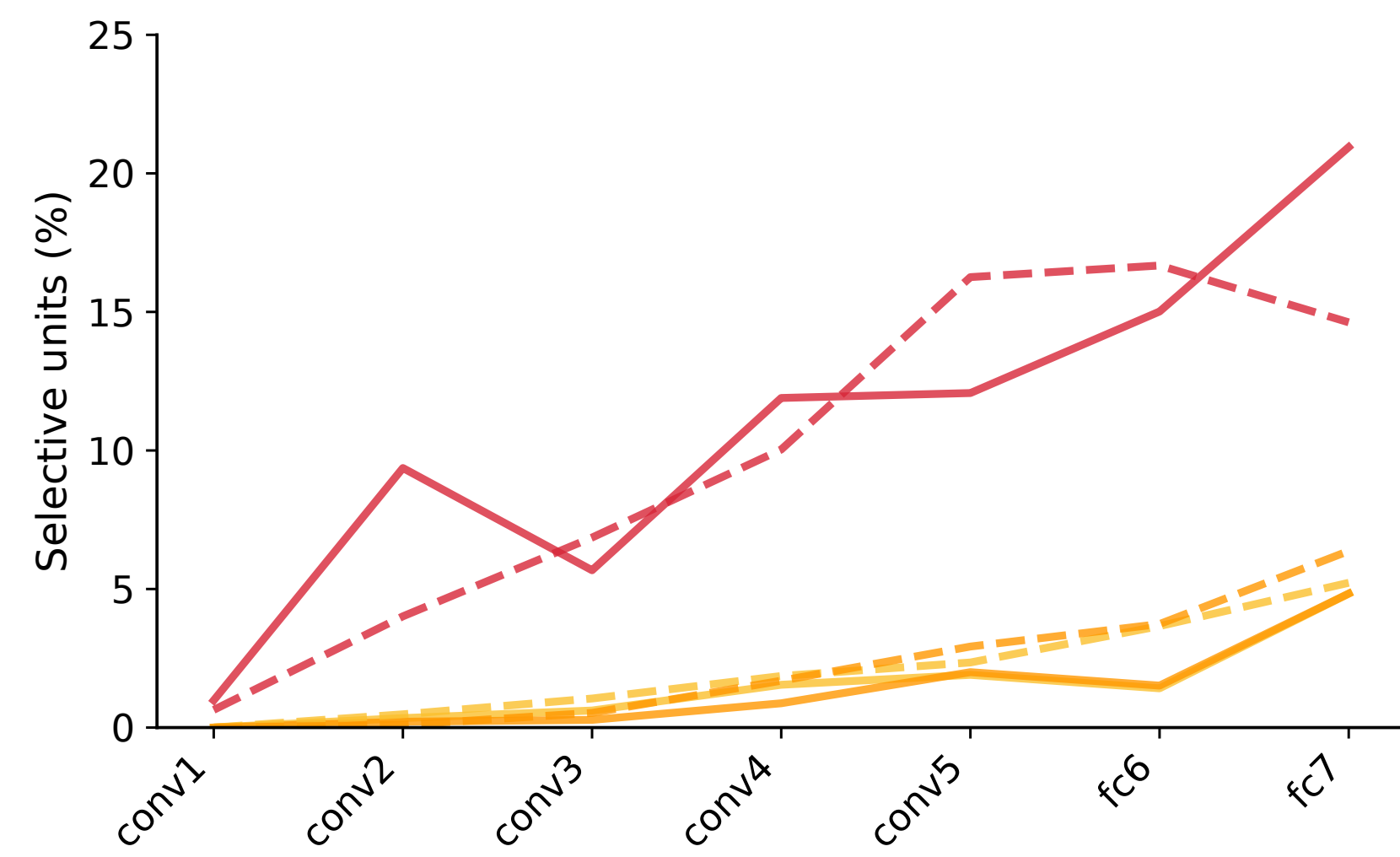
AlexNet



- Unit-wise pairwise t -tests ($p < .05$, FDR-corrected) and conjunction analysis⁵

Distinct and shared selectivity for faces and bodies in DNNs

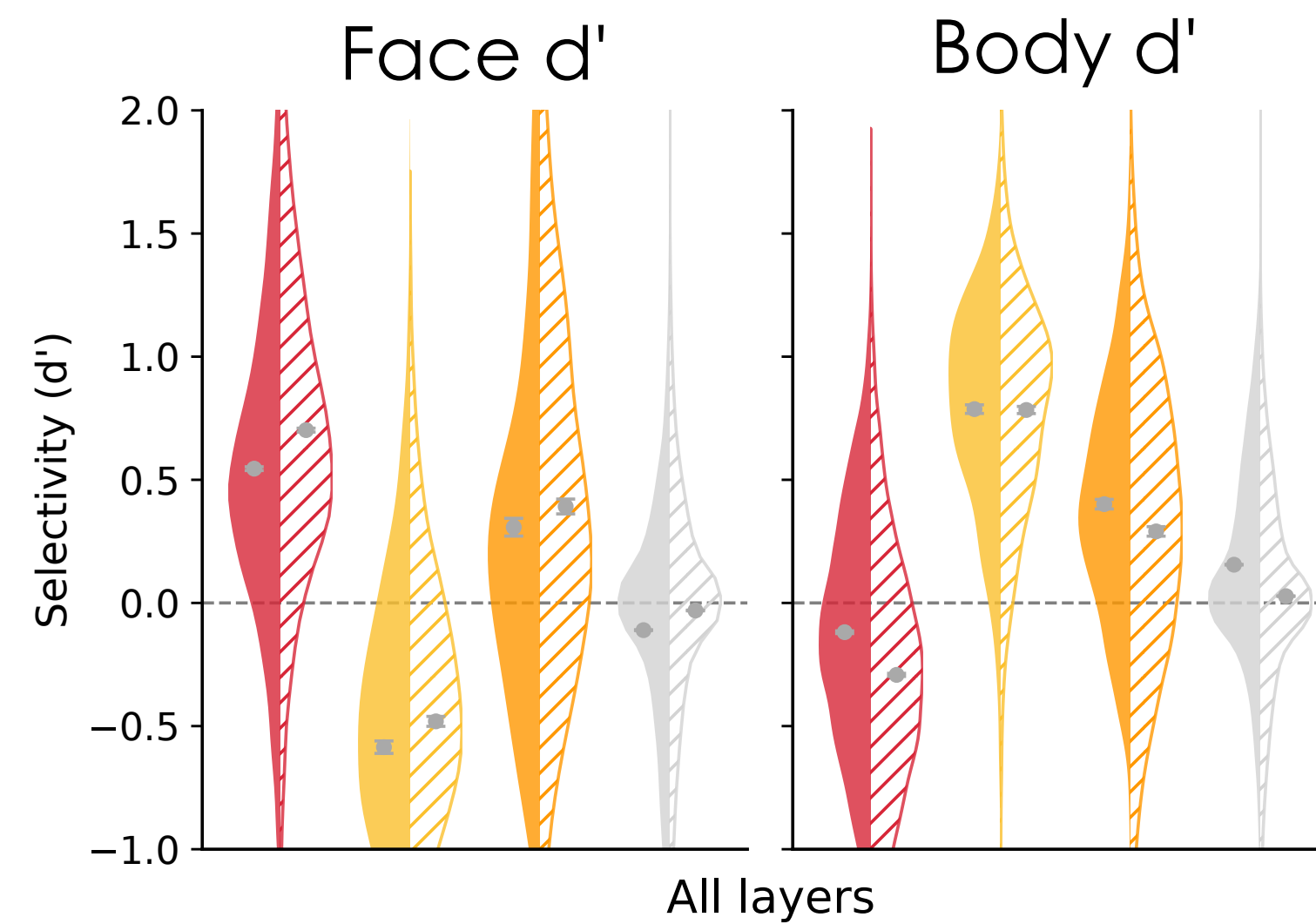
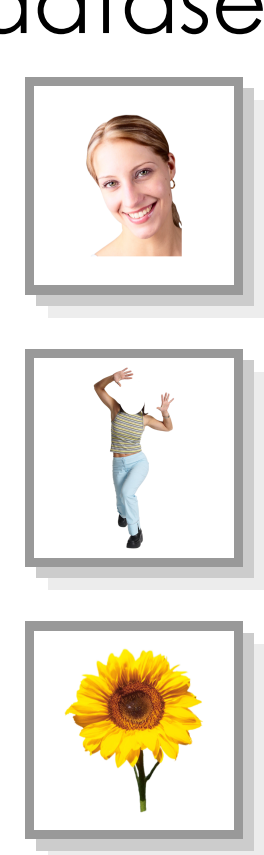
Prevalence and selectivity



Model
— AlexNet
— Supervised (Ecoset)
— AlexNet
— Self-supervised (ImageNet)

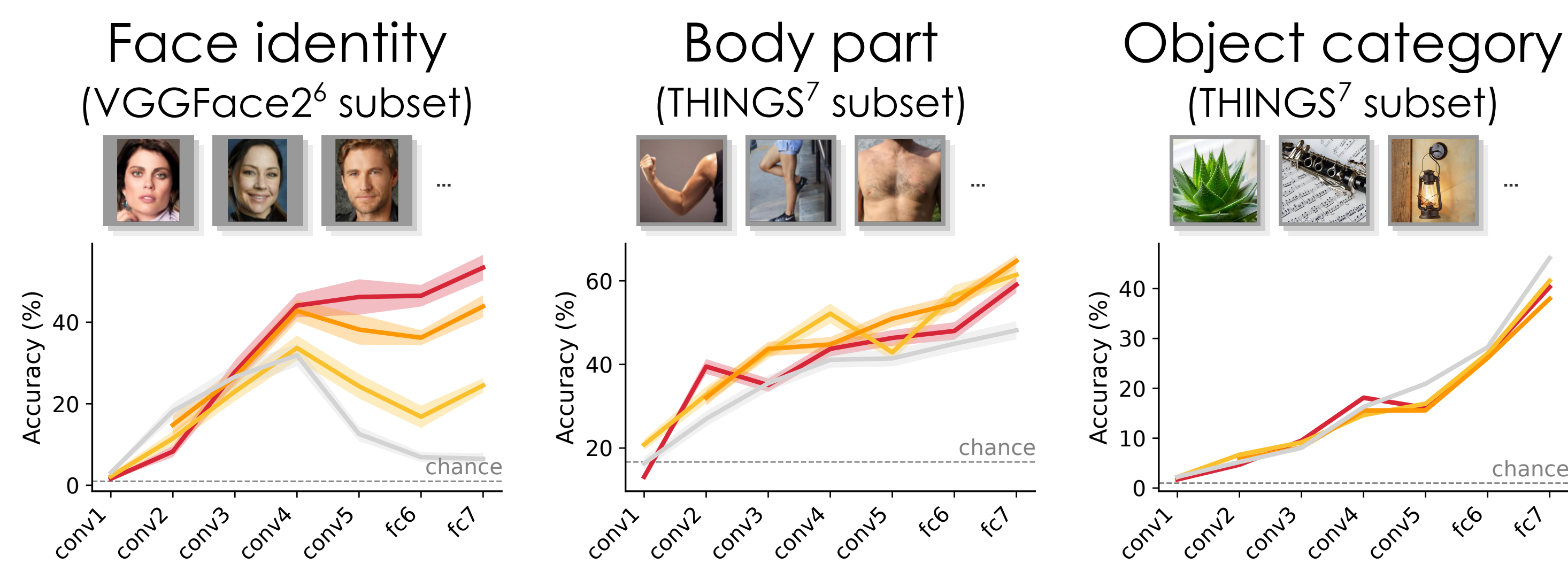
Unit type
■ Face-selective
■ Body-selective
■ Mixed-selective
■ Non-selective

Validation
dataset



- Distinct populations of face-, body-, but also mixed-selective units
- Selectivity generalizes but overlap increases

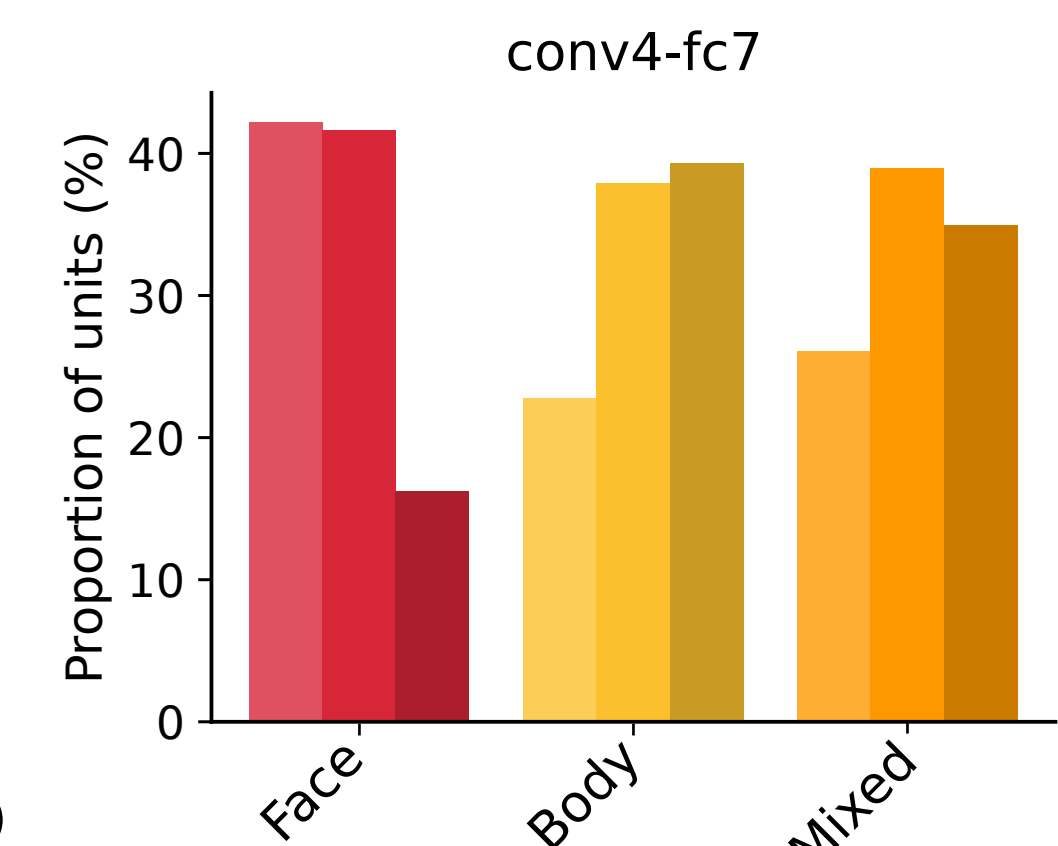
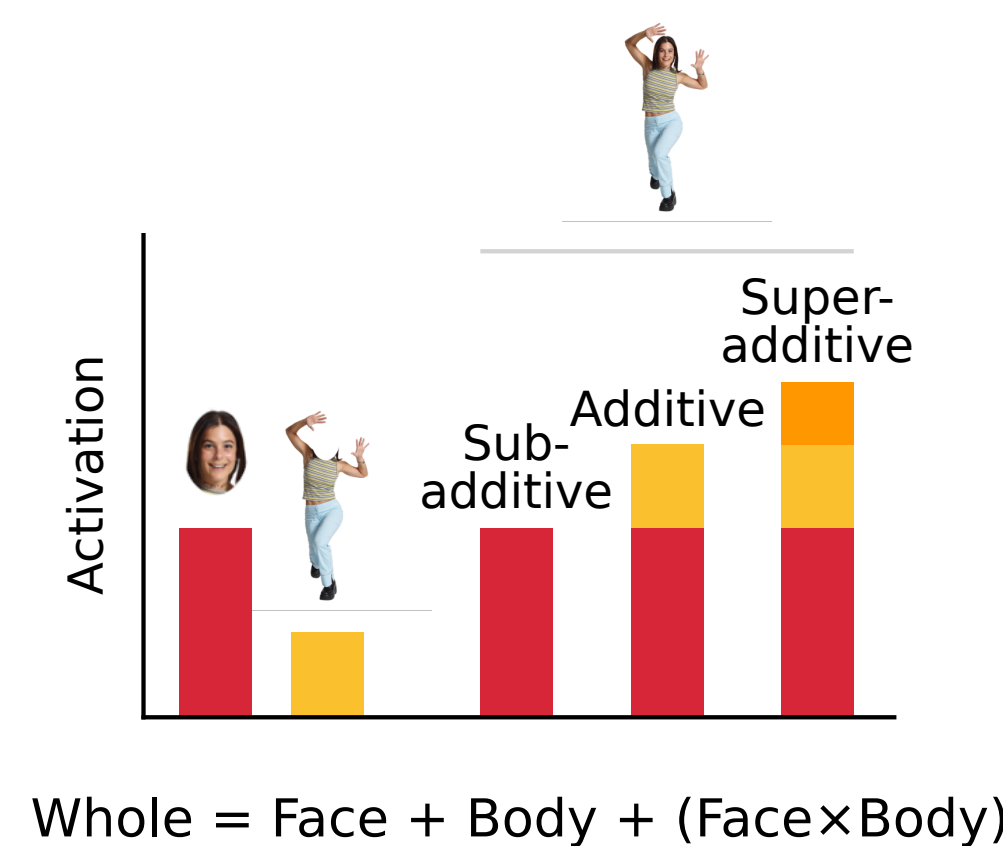
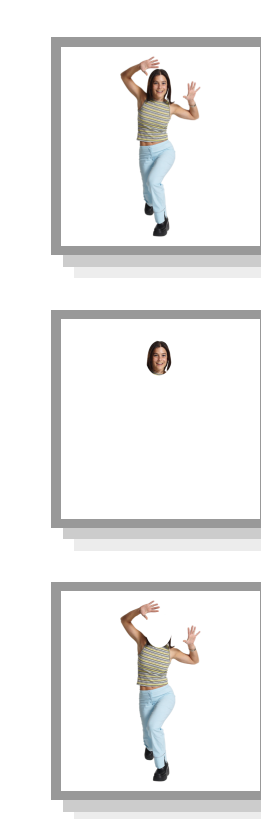
Domain-specific decoding



- Mixed-selective units encode fine-grained info from both

Face-body integration

Linear regression model



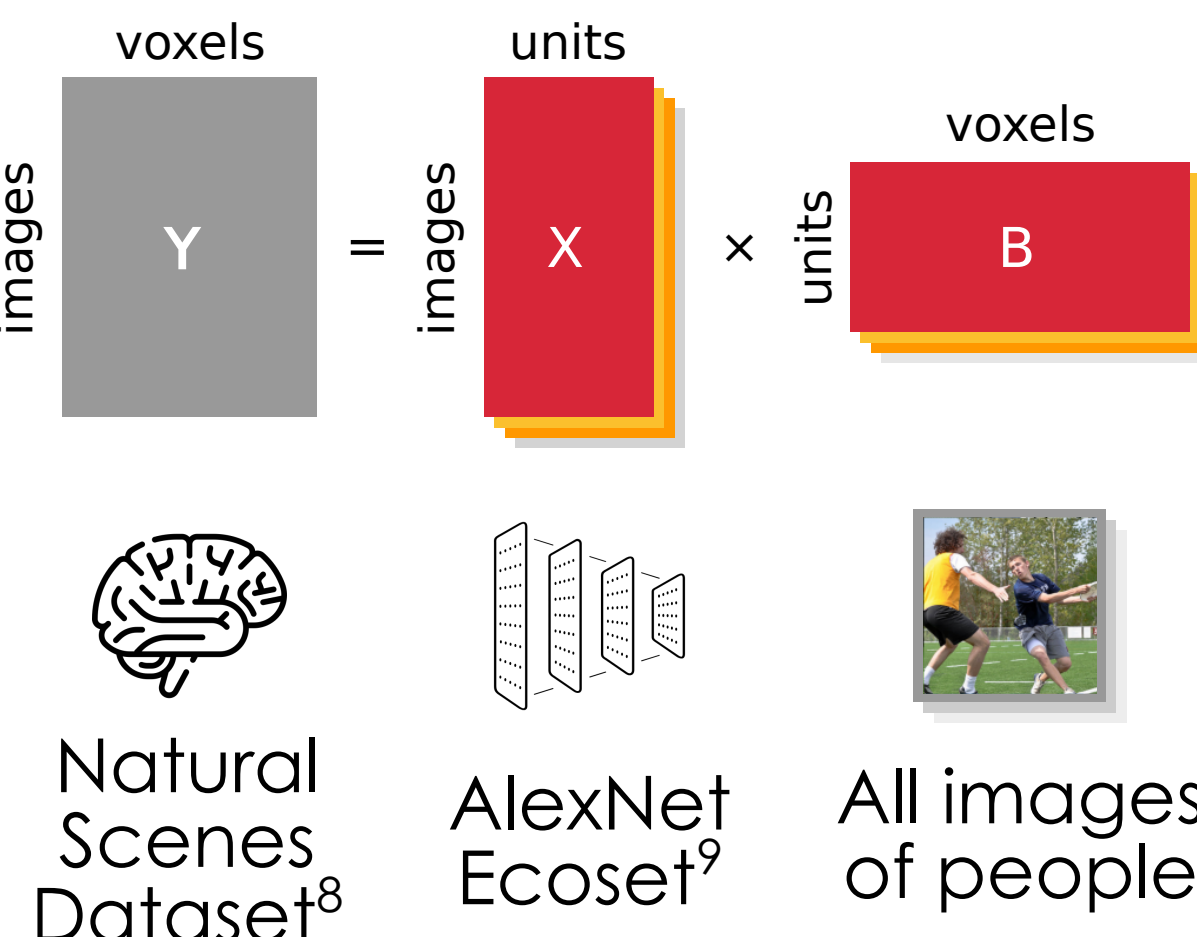
Integration
■ Sub-additive
■ Additive
■ Super-additive

- Mixed-selective units aren't special integrators

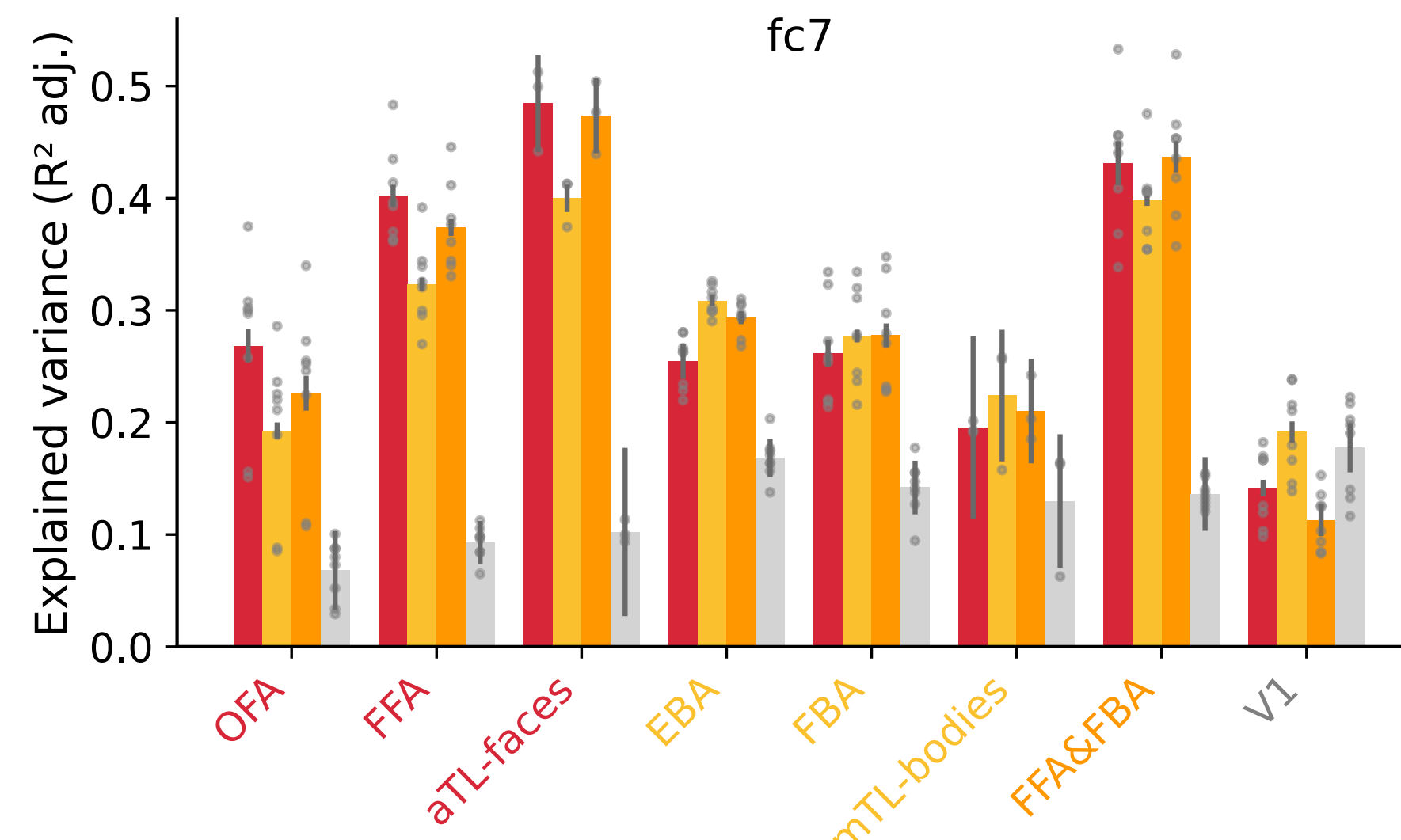
DNN units explain face- and body-selective cortical areas

fMRI encoding

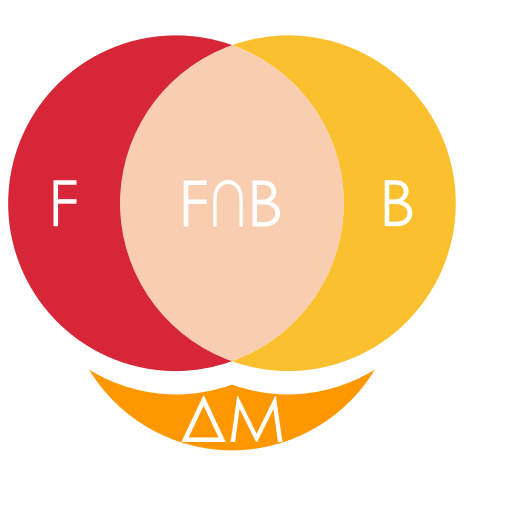
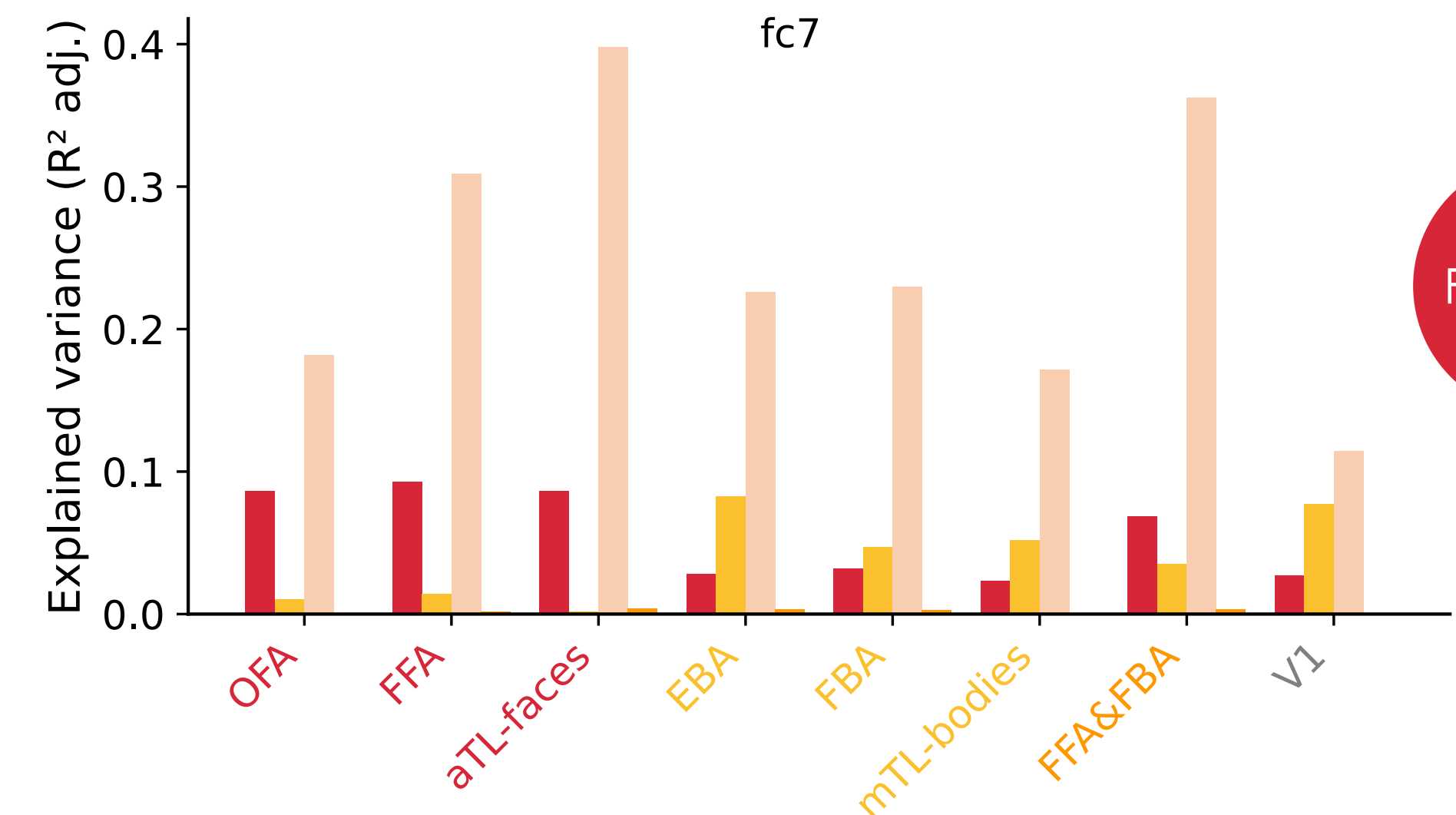
Positive ridge regression



Separate models



Variance partitioning



- DNN units best explain corresponding cortical areas

- Mixed-selective units don't explain additional variance

Evidence for strongly integrated face and body processing

- In DNNs, face and body processing are deeply intertwined, as mixed selectivity emerges early and increases progressively
- In visual cortex, face- and body-selective areas show shared tuning
- Findings support strongly integrated pathways^{10,11}

[1] Kanwisher, PNAS (2010) - [2] Schwarzlose et al., J. Neurosci. (2005) - [3] Taubert et al., Brain Struct. Funct. (2021) - [4] Stigliani et al., J. Neurosci. (2015) - [5] Prince et al., Sci. Adv. (2024) - [6] Cao et al., IEEE Conf. (2018) - [7] Hebart et al., PLOS ONE (2019) - [8] Allen et al., Nat. Neurosci. (2022) - [9] Mehrer et al., PNAS (2021) - [10] Harry et al., J. Cogn. Neurosci. (2016) - [11] Tarhan et al., Nat. Commun. (2020)